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A Multi-Source Atmospheric Refraction Framework for Geodetic Applications: ERA5 137-Level Data, Near-Surface Composites, Global Radiosonde Integration, and Real-Time Computation

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Abstract

The standard atmospheric refraction coefficient κ = 0.13, established from mid-twentieth century geodetic observations in Central Europe, has been applied globally in geodetic practice without systematic adjustment for local atmospheric conditions. While appropriate for the environments from which it was derived, this fixed value has been shown to be systematically inappropriate for non-European climates, where actual refractivity gradients can depart substantially from the ICAO standard atmosphere. This paper introduces KAPPA Arc V3, a multi-source atmospheric refraction computation framework that provides location-specific, date-specific, and time-specific values of κ derived from real measured atmospheric data for any geodetic or line-of-sight application on Earth.

The framework integrates four independent atmospheric data sources — ERA5 137-level reanalysis, ERA5-Land near-surface composites, Global Forecast System (GFS) numerical weather prediction, and NOAA IGRA2 global radiosonde profiles — across eight application-specific computation modes: line-of-sight, survey/EDM, precision levelling, radar, astronomy, marine, tunnel, and drone/UAV. A novel three-point spatial sampling methodology computes independent atmospheric profiles at observer, midpoint, and target before ensemble-averaging, improving upon single midpoint approaches for heterogeneous paths. The system provides not only κ and the combined curvature-and-refraction correction (C+R), but also the minimum required κ for geometric visibility (via iterative binary search), ray geometry at 1,000 equally-spaced path positions, detection of atmospheric thermal inversions and radio ducting conditions, modified refractivity (M-profile) for microwave propagation assessment, and supplementary radiosonde context with confidence indicators.

The framework additionally provides operator-selectable dual gradient computation comparing ordinary least-squares regression with numerical integration of the discrete refractivity profile N(z), validated across 41 globally distributed paths (95.1% excellent agreement, mean |Δκ| = 0.000126, maximum |Δκ| = 0.002287). An AI-assisted interpretation layer (Claude Sonnet 4, Anthropic 2025) translates computed κ values into mode-specific plain-language field guidance, representing the first published integration of ERA5 atmospheric reanalysis with large language model interpretation for geodetic field practice.

Operational performance is achieved through pre-extraction of regional ERA5 files to Amazon S3, automated daily updates via eight scheduled EC2 processes, and Redis result caching, reducing cold request processing times to 11–23 seconds and cached responses to 1–2 seconds. Validation against two independent published field datasets demonstrates the physical correctness of the ERA5-based approach. For the Riyadh levelling corridor (Saudi Arabia), nine-date campaign analysis spanning June–August 2012 yields κ values of 0.1925–0.2087 (campaign average κ = 0.1977), with ERA5-derived near-surface temperature gradients of −4.7 to −5.1 °C/100 m agreeing with the RCD_PLO field measurement of −5.3 °C/100 m to within 3.8–11.3% (campaign average 7.5%) — within ERA5's published near-surface accuracy of ±10–20% (Hersbach et al., 2020). This temperature gradient agreement implies equivalent κ agreement through the linear relationship $\kappa = -(R \times dN/dz \times 10^{-6})$. For the Cortes de Pallás monitoring network (Valencia, Spain), κ = 0.2279 (20 July 2023) and κ = 0.2430 (25 June 2024) are physically consistent with ERA5-based corrections by Luján et al. (2025). Three-point spatial sampling was evaluated across 41 globally distributed paths of 35–90 km, yielding higher κ than single midpoint in 26 of 41 cases (63.4%), mean $\Delta\kappa$ = +0.0006 ± 0.0037. ERA5 2m temperature agreed with 32 global IGRA2 radiosonde surface measurements with mean absolute error of 2.5 ± 1.6 °C when compared at matched UTC time and exact station coordinates. A six-comparison tropical validation against full IGRA2 radiosonde profiles at four high-density stations spanning the Atlantic, Western Pacific, and Caribbean (Barbados, Pohnpei, San Juan, Singapore) yields a mean absolute κ gap of 12.9% — within ERA5's stated ±10–20% near-surface accuracy — demonstrating that the framework's explicit wet-term computation performs correctly in tropical high-humidity environments where the wet term contributes 25–34% of total refractivity, compared to ~8% in desert conditions. The recently initiated ERA6 reanalysis (ECMWF, production commenced March 2026) at approximately twice ERA5's horizontal resolution is expected to reduce the validation gap further when data become available from late 2027. The framework is operational at arc3.kappasuite.com and is protected under Saudi patent SA 1020263556 and PCT application PCT/SA2026/050034.

Keywords: atmospheric refraction; refraction coefficient; ERA5; ERA6; geodetic levelling; EDM correction; Saudi Arabia; line-of-sight visibility; radiosonde; ducting; thermal inversion; three-point spatial sampling; KAPPA Arc V3; real-time atmospheric correction

1. Introduction

Electromagnetic waves propagating through the lower troposphere do not travel in straight lines. The vertical gradient of atmospheric refractivity causes the ray path to curve, introducing systematic errors in all terrestrial geodetic measurements that depend on the geometric relationship between observer and target. In distance measurement by electronic distance meters (EDM), in vertical angle observations by total stations, in optical levelling, in radar propagation, and in any line-of-sight application, the magnitude of this refraction-induced error is characterised by the atmospheric refraction coefficient κ, defined as the ratio of the mean radius of the Earth R to the radius of curvature ρ of the refracted ray path.

The value κ = 0.13, derived from empirical geodetic observations in Central Europe during the mid-twentieth century and codified by Jordan and Eggert (1939) and Bomford (1952), has served as the global standard for refraction correction in geodetic practice for over seven decades. It has been widely applied in geodetic practice regardless of geographic location or atmospheric conditions, and is incorporated in atmospheric correction procedures recommended in geodetic textbooks and national survey standards (Bomford, 1952; Jordan and Eggert, 1939).

κ = 0.13 is a historically derived empirical approximation, not a value derived from first-principles atmospheric physics. Substantial evidence has accumulated that it does not adequately represent atmospheric refraction across the range of climatic environments in which geodetic surveys are conducted, with ERA5-derived κ values consistently exceeding 0.13 even in European locations (see Section 5.1). Simultaneous reciprocal vertical angle measurements in alpine terrain have demonstrated diurnal and seasonal variability of κ from −0.05 to 0.40 (Hirt et al., 2010), confirming that the standard fixed value is inadequate even in European environments. The RCD_PLO project (General Commission for Survey, 2018) processed 1,200,000+ precision levelling records across Saudi Arabia and derived a vertical temperature gradient of −5.3 °C/100 m for summer conditions in the Riyadh corridor — more than eight times stronger than the ICAO standard underlying κ = 0.13. Luján et al. (2025) demonstrated that ERA5-based refraction corrections substantially outperformed standard meteorological corrections at a geodetic monitoring network in Valencia, Spain.

These studies collectively demonstrate that the departure of actual atmospheric conditions from the ICAO standard is not a minor perturbation but a systematic, geographically and seasonally structured phenomenon. In hot desert environments such as the Arabian Peninsula, the near-surface temperature gradient can be 7–8 times stronger than the ICAO standard, producing refraction corrections that are 130–140% larger than the standard value would suggest. In coastal and humid tropical environments, high water vapour content increases the wet term contribution to refractivity significantly beyond what the standard value accounts for. Even in European alpine terrain, diurnal heating and cooling cycles produce substantial excursions from the standard value.

The companion paper to the present work (Alkalali, 2026a, GEOG-D-26-00098, under review) established the ERA5 137-level methodology for deriving κ from reanalysis atmospheric profiles, validated for a 493 km line-of-sight baseline in the Caucasus mountain range. The present paper introduces KAPPA Arc V3 — an operational multi-source framework that extends this methodology to eight geodetic application modes, covering the full range of terrestrial geodetic measurement types from precision levelling to long-range radar propagation. The central purpose of KAPPA Arc V3 is to provide geodetic practitioners with the correct atmospheric refraction coefficient for their specific location, application, date, and time — replacing the single fixed global constant with a physically grounded, measurement-based computation that can be performed operationally before or during fieldwork.

The key scientific and engineering contributions of this work are: (1) a multi-source atmospheric data architecture that assigns the optimal ERA5, ERA5-Land, GFS, or radiosonde data source based on application mode and path geometry; (2) a novel three-point spatial sampling methodology that computes independent atmospheric profiles at observer, midpoint, and target to account for horizontal atmospheric variation along the path;

(3) complete atmospheric output including not only κ and C+R corrections but also ray geometry, thermal inversion and radio ducting detection, modified refractivity profiles, and radiosonde confidence indicators; (4) an operational infrastructure delivering results in 11–23 seconds for any location on Earth; and (5) comprehensive validation against two independent published field datasets using nine-date campaign-period analysis, 41-path three-point sampling evaluation, and 16-station global radiosonde comparison; (6) an operator-selectable dual gradient computation method comparing OLS regression with numerical integration, validated across 41 globally distributed paths (95.1% excellent agreement, mean $|\Delta\kappa| = 0.000126$, maximum $|\Delta\kappa| = 0.002287$, zero DIVERGENT cases); and (7) an AI-assisted natural language interpretation layer that translates computed κ values into mode-specific field guidance, representing the first published integration of ERA5 reanalysis with large language model interpretation for geodetic practice.

2. Theoretical Background

2.1 The Atmospheric Refraction Coefficient

The atmospheric refraction coefficient κ quantifies the curvature of an electromagnetic ray path propagating through the lower troposphere relative to the curvature of the Earth's surface. It is formally defined as:

$$\kappa = R / \rho$$

where R is the mean radius of the Earth (6,371,000 m) and ρ is the radius of curvature of the refracted ray. For a horizontally stratified atmosphere, κ is related to the vertical gradient of the atmospheric refractivity N through:

$$\kappa = -(R / 2) \times 10^{-6} \times dN/dz \quad \text{[classical two-ray derivation]}$$

where z is geometric altitude above the geoid. Note on sign convention: the formula above follows the classical two-ray geometric derivation (Bomford, 1952) and includes a factor of 1/2. An alternative single-ray convention, $\kappa = -(R \times dN/dz \times 10^{-6})$, is used in some geodetic literature and is the convention implemented in KAPPA V3 (Section 3.4). The standard reference value κ = 0.13 also uses the single-ray convention; all comparisons in this paper are therefore internally consistent. The two conventions differ by a factor of 2 in κ for the same dN/dz. The atmospheric refractivity N (dimensionless × 10^6) is computed using the Smith-Weintraub formula (Smith and Weintraub, 1953):

$$N = 77.6 \times P/T + 373000 \times e/T^2$$

where P is atmospheric pressure in hPa, T is temperature in Kelvin, and e is the partial pressure of water vapour in hPa. The first term (77.6 × P/T) is the dry term, dominated by atmospheric pressure and temperature. The second term ($373000 \times e/T^2$) is the wet term, reflecting the disproportionately large contribution of water vapour to refractivity relative to its mass fraction in the atmosphere. In dry desert environments such as the Arabian Peninsula, relative humidity of 5–15% means the wet term typically contributes 8–12% of total refractivity; in humid tropical and maritime environments, relative humidity exceeding 80% can raise the wet term contribution to 20–30% of total N. KAPPA Arc V3 computes both terms explicitly at each ERA5 model level, ensuring accurate refractivity computation across the full range of humidity conditions encountered globally.

Water vapour partial pressure e is derived from ERA5 specific humidity q at each model level as:

$$e(k) = q(k) \times P(k) / (0.622 + 0.378 \times q(k))$$

This formulation ensures thermodynamic consistency between the specific humidity field retrieved from ERA5 and the vapour pressure used in the Smith-Weintraub formula, avoiding the approximation errors introduced by simpler conversion methods.

The refraction coefficient κ governs the magnitude of geodetic corrections through several application-specific formulae. For combined curvature and refraction correction in geodetic levelling and EDM:

$$C+R = d^2(1 - 2\kappa) / (2R)$$

where d is the sight or path distance in metres. For angular refraction correction of observed zenith angles:

$$\delta z = \kappa \times d / (2R) \text{ [radians]}$$

The sensitivity of these corrections to the value of κ is linear. A departure of $\Delta\kappa = 0.07$ from the standard value — typical for Saudi Arabian summer conditions — produces a proportional error in the C+R correction that grows quadratically with path length. For a 500 m levelling sight, this corresponds to approximately 2.7 mm of systematic error per setup — an amount that accumulates significantly over kilometre-scale levelling networks.

The propagation of ERA5 temperature uncertainty to κ uncertainty can be formalised through the Smith-Weintraub equation. The sensitivity of refractivity N to temperature T at constant pressure and humidity is $\partial N/\partial T = -77.6P/T^2 - 2\times373000\times e/T^3$, yielding approximately −0.89 N/K under Saudi Arabian conditions ($P \approx 944$ hPa, $T \approx 311$ K, $e \approx 5.4$ hPa) and −1.58 N/K under tropical maritime conditions ($P \approx 1007$ hPa, $T \approx 298$ K, $e \approx 25$ hPa). For an OLS gradient computed over n ERA5 levels spanning altitude range Δz, the resulting κ uncertainty is: $\sigma_\kappa \approx R \times (|\partial N/\partial T| \times \sigma_T) / \sqrt{[\Delta z^2(n^2-1)/12]} \times 10^{-6}$. With $\sigma_T = 2.5$°C (the mean ERA5 surface temperature error from Section 4.3), $\Delta z = 500$ m, and $n \approx 25$ ERA5 levels, this yields $\sigma_\kappa \approx \pm0.004$ under Saudi conditions and ±0.007 under tropical conditions. These values represent 2–3% of the typical κ values derived in this study (0.19–0.27), confirming that ERA5 temperature uncertainty is a secondary contributor to total κ uncertainty — the dominant source remains ERA5's representation of the refractivity gradient shape, as quantified by the 7.5% (Saudi) and 12.9% (tropical) validation gaps.

2.2 The Modified Refractivity and Ducting

For microwave and radar propagation applications, the relevant atmospheric quantity is the modified refractivity M, which accounts for the Earth's curvature in the ray path analysis:

$$M(z) = N(z) + (z / R) \times 10^6$$

The vertical gradient of M determines the propagation regime. When $dM/dz > 0$ (the normal condition), rays curve downward less steeply than the Earth's surface and propagation is standard. When $dM/dz \approx 0$, rays follow the Earth's surface (standard refraction). When dM/dz

< 0 in a layer, rays are trapped within that layer — a condition known as radio ducting or super-refraction — which can cause anomalous propagation at distances far beyond the normal radio horizon, affecting radar performance, microwave link reliability, and geodetic EDM measurements at low elevation angles. KAPPA Arc V3 computes M at each sampled altitude and provides the M-profile for assessment of ducting conditions in the Radar application mode.

2.3 Effective Earth Radius and Ray Geometry

A useful equivalent formulation for ray path analysis replaces the curved ray on a spherical Earth with a straight line on a sphere of enlarged effective radius:

$$R_{eff} = R / (1 - \kappa)$$

On this effective sphere, the refracted ray travels in a straight line between observer and target. KAPPA Arc V3 uses this formulation to compute the ray geometry at 1,000 equally-spaced positions along the observer-target chord. At each position i/1000 of the total path length, the height of the chord above the effective sphere surface is:

$$h(i) = h_A + (h_B - h_A)(i/N) - (D^2/2R_{eff})(i/N)(1 - i/N)$$

where h_A and h_B are observer and target elevations, D is path length, and $n_{pos} = 1{,}000$ is the number of path positions. The minimum of h(i) across all 1,000 positions gives the minimum ray clearance height — the closest approach of the refracted ray to the Earth's surface. If this minimum is non-negative at all intermediate terrain points, the target is geometrically visible from the observer under the computed κ.

2.4 ERA5 Reanalysis and Model Level Structure

ERA5, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) using the Integrated Forecasting System with 4D-Var data assimilation, represents the atmospheric state on 137 hybrid sigma-pressure model levels defined by coefficients a(k) and b(k):

$$p(k) = a(k) + b(k) \times p_{surface}$$

These coefficients define a terrain-following coordinate system near the surface that transitions to pure pressure coordinates at high altitudes. The 137 levels range from the surface (level 137, pressure approximately equal to surface pressure) to approximately 0.01 hPa (level 1, stratopause), with the highest vertical resolution concentrated in the planetary boundary layer where atmospheric refractivity gradients are most pronounced. Approximately 20–30 model levels lie below 1 km above ground level (depending on surface elevation; ~20–25 at 600 m ASL in the Saudi corridor), compared to 2–3 levels in ERA5's 37-pressure-level product — a factor of approximately 8–10 improvement in boundary layer vertical resolution that is essential for accurate refractivity gradient computation in the near-surface layer most relevant to geodetic applications.

ERA5-Land (Muñoz-Sabater et al., 2021) is a companion product that combines ERA5 output with a high-resolution land surface model, providing 2 m temperature, 2 m dewpoint temperature, skin temperature, and surface pressure at 0.1° horizontal resolution. The skin temperature — the radiative surface temperature — is particularly important for geodetic levelling applications, where the temperature at ground level determines the bottom boundary condition for the near-surface temperature gradient. In desert environments, daytime skin temperatures can reach 60–70°C, creating extreme near-surface lapse rates that ERA5 at its coarser resolution may partially underestimate. KAPPA Arc V3 uses ERA5-Land skin temperature as a 0 m anchor in the levelling and survey composite profile, merging into ERA5 L137 levels above approximately 10 m.

The recently initiated ERA6 reanalysis (ECMWF, production commenced March 2026) is expected to provide approximately twice the horizontal resolution of ERA5 (~14 km vs ~31 km) with explicitly improved representation of near-surface quantities (Copernicus Climate Change Service, 2026). ERA6 data are expected to become available progressively from late 2027. KAPPA's data-source-agnostic architecture — protected under Claim 3 of patent SA 1020263556, covering 'any source of measured vertical atmospheric profiles' — enables direct ERA6 integration without framework modification when data become available.

3. The KAPPA Arc V3 Framework

3.1 Framework Purpose and Design Philosophy

KAPPA Arc V3 is designed to provide geodetic practitioners with the physically correct atmospheric refraction coefficient for their specific measurement situation — replacing a single global constant with a computation grounded in real atmospheric data. The framework addresses three practical needs that the standard fixed value cannot meet: (1) geographic variation — atmospheric conditions differ fundamentally between Central Europe, the Arabian Peninsula, tropical regions, and alpine terrain; (2) temporal variation — conditions change hour by hour with solar heating, moisture advection, and synoptic weather systems; and (3) application specificity — optical levelling at rod heights of 0.3–2.5 m experiences different atmospheric conditions than long-range radar propagation at several hundred metres above ground.

The design philosophy prioritises operational usability over theoretical completeness. Rather than requiring users to understand atmospheric physics or process raw reanalysis data, KAPPA Arc V3 accepts the same input parameters a surveyor would naturally know — observer and target coordinates, elevation, date, time, and application type — and returns all refraction-related quantities needed for measurement correction, visibility assessment, and quality control, within a processing time compatible with field planning workflows.

3.2 Eight Application Modes and Data Source Assignment

KAPPA Arc V3 implements eight application-specific computation modes, each assigned the atmospheric data source most appropriate for the physical characteristics of that measurement type. The assignment is governed by two principles: the altitude range of the atmospheric layer traversed by the line of sight, and the temporal currency requirements of the application.

Mode	Primary Source	Secondary Source	Physical Basis
Line-of-Sight (LOS)	ERA5 L137 (137 levels)	IGRA2 radiosonde	Full tropospheric column; paths typically >10 km
Radar / Microwave	ERA5 L137 (137 levels)	ERA5-Single (BLH)	Ducting assessment requires boundary layer resolution
Astronomy	ERA5 L137 (137 levels)	ERA5-Single (TCWV)	Zenith wet delay from total column water vapour
Survey / EDM	ERA5-Land + L137	IGRA2 radiosonde	Near-surface path; ERA5-Land 0.1° for surface layer
Precision Levelling	ERA5-Land + L137	IGRA2 radiosonde	Rod heights 0.3–2.5 m; skin temperature anchor at 0 m
Marine	ERA5-Land + L137	ERA5-Single (SST)	Sea surface temperature governs marine boundary layer
Tunnel	ERA5-Land + L137	—	Enclosed near-surface; no full column required
Drone / UAV	GFS Real-Time	—	Operational forecast for current or future conditions

Table 1. Application mode to data source assignment in KAPPA Arc V3. BLH: boundary layer height; TCWV: total column water vapour; SST: sea surface temperature.

A note on the Tunnel mode is warranted. Geodetic total station surveys inside tunnels involve sight lines through the enclosed tunnel atmosphere. However, ERA5 external atmospheric profiles remain the appropriate data source for two reasons. First, tunnel atmospheric conditions are strongly coupled to the external atmosphere through ventilation portals and thermal conduction; temperature and humidity gradients inside operational tunnels during survey campaigns are empirically close to external near-surface conditions in the absence of active ventilation anomalies. Second, no operational data source provides real-time internal tunnel atmospheric profiles; ERA5-Land near-surface composites representing external boundary conditions constitute the best available physically grounded estimate. Users surveying in tunnels with active ventilation or significant traffic-induced thermal anomalies should treat KAPPA V3 outputs as first-order estimates and supplement with in-situ measurements where First Order accuracy is required.

The LOS mode uses the full ERA5 137-level profile to characterise the entire atmospheric column traversed by long-range line-of-sight observations, where the ray path may span altitudes from near-surface to several hundred metres above ground over horizontal distances of tens to hundreds of kilometres. The Radar mode adds ERA5-Single boundary layer height to identify the vertical extent of the ducting-susceptible layer. The Astronomy mode incorporates total column water vapour for zenith wet delay correction relevant to optical and radio astronomy. Survey and Levelling modes use the ERA5-Land composite profile to improve near-surface accuracy at the altitudes most relevant to ground-based survey instrumentation. The Marine mode uses ERA5 sea surface temperature to better characterise the marine boundary layer, where air-sea temperature difference drives near-surface refractivity gradients. The Drone mode uses GFS forecast data, enabling computation for future observation times to support mission planning.

3.3 Three-Point Spatial Sampling

The standard approach to atmospheric refraction correction evaluates a single atmospheric profile at the geometric midpoint of the observation path, implicitly assuming that this profile is representative of conditions along the entire path. This assumption is well-supported for short paths within horizontally homogeneous atmospheric environments, but becomes increasingly problematic for longer paths and for observations crossing boundaries between atmospheric regimes — coastal-to-inland transitions, valley-to-ridge ascents, or paths spanning large horizontal temperature gradients.

KAPPA Arc V3 implements three-point spatial sampling along the observation path. Multi-point atmospheric sampling along propagation paths is established in GPS tropospheric delay modelling and radio propagation (Davis et al., 1985; Niell, 1996); the specific novelty here is the application of this principle to ERA5 L137 native model-level profiles for geodetic refraction coefficient computation. Independent atmospheric profiles are extracted at three geographic positions — the observer location A (lat_A, lon_A, elev_A), the geometric midpoint M (the midpoint of the great circle arc from A to B), and the target location B (lat_B, lon_B, elev_B). At each position, the refractivity gradient dN/dz is computed from the local ERA5 profile using least-squares regression across the atmospheric levels intersected by the estimated line-of-sight altitude at that horizontal position. The three gradients are then ensemble-averaged:

$$(dN/dz)\_path = [(dN/dz)\_A + (dN/dz)\_M + (dN/dz)\_B] / 3$$

This three-point average represents a first-order numerical quadrature of the true path-integrated refractivity gradient. Rather than assuming that atmospheric conditions at the geometric midpoint are representative of the entire path, three-point sampling explicitly characterises conditions at the observer location, midpoint, and target location before ensemble averaging. Three-point sampling is physically more representative than single midpoint for all observation paths — the methodology advantage becomes numerically significant for paths crossing horizontal atmospheric gradients such as coastal boundaries, orographic features, and thermal regime transitions. To quantify the difference from single midpoint sampling, 41 globally distributed paths of 35–90 km were evaluated using the same ERA5 L137 data pipeline for both methods, with κ_mid computed from the midpoint profile alone and κ_3pt from the three-point ensemble average. Three-point sampling yielded higher κ in 26 of 41 cases (63.4%), with mean $\Delta\kappa = +0.0006 \pm 0.0037$ (see Section 5.2 and Supplementary S2 for full results).

3.4 ERA5 L137 Profile Extraction and κ Computation

For each of the three sampling points, the following computational sequence is executed. The two ERA5 six-hourly analysis times bracketing the requested observation time are identified. For each bracketing time, the pre-extracted regional ERA5 L137 NetCDF file is retrieved from Amazon S3. Temperature T(k) and specific humidity q(k) are read at all model levels k = 75 to 137 (the 63 levels from approximately the surface to 20 km altitude). Surface pressure p_surface is retrieved from the ERA5 surface pressure field.

Pressure at each model level is computed from ERA5 hybrid sigma-pressure coefficients:

$$p(k) = a(k) + b(k) \times p\_surface \; [hPa]$$

Geometric height z(k) at each level is derived from the ERA5 geopotential height field φ(k):

$$z(k) = \varphi(k) / g_0 \; [metres], \text{ where } g_0 = 9.80665 \; m/s^2$$

Water vapour partial pressure at each level is computed from specific humidity:

$$e(k) = q(k) \times p(k) / (0.622 + 0.378 \times q(k)) \; [hPa]$$

Atmospheric refractivity at each level is computed using the Smith-Weintraub formula (Smith and Weintraub, 1953):

$$N(k) = 77.6 \times p(k)/T(k) + 373000 \times e(k)/T(k)^2$$

The refractivity gradient dN/dz is estimated by ordinary least-squares linear regression of N(k) against z(k) across the subset of model levels whose geometric height falls within the atmospheric slice bounded by the observer elevation and target elevation at the sampling point. Linear temporal interpolation is applied between the refractivity gradients computed from the two bracketing ERA5 analyses, weighted by temporal proximity to the requested observation time.

The atmospheric refraction coefficient is then derived as:

$$\kappa = -(R \times dN/dz \times 10^{-6}), \text{ where } R = 6,371,000 \; m$$

As a concrete example from a computation for the Riyadh corridor on 15 July 2025 at 10:00 UTC: at the observer location (24.60°N, 46.80°E, 600 m elevation), ERA5 returned T = 311.51 K (38.36°C), RH = 8.1%, e = 5.404 hPa. Applying the Smith-Weintraub formula at the observer level (p = 943.5 hPa): $N = 77.6 \times 943.5/311.51 + 373000 \times 5.404/311.51^2 = 235.07 + 20.74 = 255.81$. The dry term (235.07, 92% of total) confirms low humidity; the wet term (20.74, 8% of total) is non-negligible even in desert conditions. The computed dN/dz = −29.96 N/km yields κ = 0.1909, compared to κ = 0.13 under standard assumptions — a 47% departure.

Independent replication of any KAPPA V3 result is possible using freely available ERA5 data and the published methodology. The computation requires: (1) ERA5 model-level temperature ('t'), specific humidity ('q'), and surface pressure ('sp') downloaded from the Copernicus Climate Data Store for the ERA5 native model-level product ('reanalysis-era5-complete', levtype='ml'); (2) pressure at each level k computed as p(k) = a(k) + b(k) x p_surface, using ECMWF hybrid coefficients tabulated in the ERA5 documentation; (3) geometric altitude z(k) = phi(k)/g0; (4) vapour pressure e(k) = q(k) x p(k) / (0.622 + 0.378 x q(k)); (5) Smith-Weintraub refractivity N(k) = 77.6 x p(k)/T(k) + 373000 x e(k)/T(k)2; and (6)

OLS regression of N against z over the 0-500 m window to yield dN/dz, from which kappa = -(R x dN/dz x 10-6). For the Riyadh corridor example above (24.60N, 46.80E, 15 July 2025, 10:00 UTC), this procedure applied to publicly downloaded ERA5 data yields the values reported. Any researcher with Copernicus CDS access can replicate this computation directly from raw ERA5 files without use of the KAPPA API.

3.5 ERA5-Land Near-Surface Composite Profile

For application modes involving near-surface observations (Survey, Levelling, Marine, Tunnel), the ERA5-Land composite profile is used to improve accuracy in the critical 0–10 m layer. The composite is constructed as follows. ERA5-Land fields for skin temperature T_{skin} (the radiative surface temperature at 0 m above ground), 2 m temperature T_{2m}, 2 m dewpoint temperature T_{d2m}, and surface pressure $p_{surface}$ are retrieved for the sampling point and observation time. The skin temperature anchors the refractivity profile at ground level:

$$N_{skin} = 77.6 \times p_{surface} / T_{skin} + 373000 \times e_{skin} / T_{skin}^2$$

where e_{skin} is derived from T_{d2m} as $e_{skin} = 6.112 \times 10^{(7.5 \times Tc_{d2m} / (237.3 + Tc_{d2m}))}$ with $Tc_{d2m} = T_{d2m} - 273.15$. The 2 m layer refractivity is computed similarly from T_{2m} and T_{d2m}. These two near-surface levels (0 m and 2 m) are prepended to the ERA5 L137 profile above approximately 10 m, providing a continuous composite profile from ground level through the full tropospheric column. This composite is particularly important for precision levelling, where rod readings at 0.3–2.5 m above ground are most sensitive to the steepest part of the near-surface temperature gradient.

3.6 Binary Search for Required Refraction Coefficient

In addition to computing the atmospheric κ from measured profiles, KAPPA Arc V3 implements an iterative binary search algorithm to determine the minimum κ required for geometric visibility between observer and target — that is, the minimum κ at which the refracted ray clears all terrain between the two points without intersecting the ground surface. The algorithm operates as follows. A search interval $[\kappa_{min}, \kappa_{max}] = [-0.5, 2.0]$ is initialised, spanning the physically plausible range of atmospheric refraction coefficients from super-refraction to extreme ducting. At each iteration, the midpoint κ_{mid} is computed, the effective Earth radius $R_{eff} = R/(1 - \kappa_{mid})$ is updated, the minimum ray clearance height across 1,000 equally-spaced path positions is evaluated, and the search interval is halved based on the sign of the minimum clearance. The search converges to machine precision ($\sim 10^{-15}$) within 50 iterations. The required-κ output enables direct comparison with the atmospheric κ: if the atmospheric κ exceeds the required κ, the target is visible; if not, there is a deficit that quantifies the refraction shortfall for visibility.

3.7 Atmospheric Anomaly Detection

KAPPA Arc V3 scans the ERA5 vertical temperature profile at each sampling point to detect two categories of atmospheric anomalies that significantly affect geodetic measurement quality.

Thermal inversion detection: the vertical temperature profile retrieved from ERA5 is scanned for altitude intervals in which temperature increases with altitude ($dT/dz > 0$). Such temperature inversions are significant in geodetic contexts because they produce refractivity gradients that can be substantially larger than under normal lapse rate conditions, causing enhanced ray curvature that systematic refraction corrections based on standard κ cannot account for. When an inversion is detected, KAPPA Arc V3 reports the base altitude, top altitude, depth, and temperature gradient of each inversion layer.

Radio ducting detection: the M-profile (modified refractivity, $M = N + z/R \times 10^6$) is computed at each ERA5 level. A layer with negative dM/dz constitutes a ducting layer, in which microwave and radar rays are trapped, propagating anomalously beyond the normal radio horizon. KAPPA Arc V3 identifies such layers and reports the trapping layer altitude for Radar mode computations. In the example response for the Riyadh corridor (15 July 2025, 10:00 UTC), M increased monotonically from 349.98 at 600 m to 356.33 at 650 m ($dM/dz > 0$), indicating no ducting — physically expected under the strongly superadiabatic desert boundary layer conditions present on that day, which promote normal dispersion rather than trapping.

3.8 Radiosonde Integration and Confidence Indicators

For each KAPPA Arc V3 computation, the system identifies the nearest active radiosonde station from the pre-processed NOAA IGRA2 archive of 889 globally active stations (January 2025 onward). The most recent sounding within ±48 hours of the requested observation time is retrieved, providing an independent observational characterisation of the atmospheric column near the measurement path. The radiosonde sounding is presented as supplementary context alongside the ERA5-derived κ — it does not modify the primary computation, but serves as a quality indicator.

Station proximity is colour-coded as a confidence indicator: green for stations within 100 km of the path midpoint (highly representative of local conditions), amber for 100–300 km (use with caution — may represent different terrain or micro-climate), and red for stations beyond 300 km (may not represent local conditions — ERA5 reanalysis provides better spatial coverage than any individual distant sounding). For the Riyadh corridor, the nearest active radiosonde station is King Fahd Airport, Dammam (SAM00040417, 26.45°N, 49.81°E), located 362.9 km from the path midpoint — correctly classified as red, reflecting the sparse radiosonde network density in the Arabian Peninsula.

3.9 Dual Gradient Computation: OLS Regression and Numerical Integration

The refractivity gradient dN/dh described in Section 3.4 is derived by ordinary least-squares (OLS) linear regression, which assumes a linear variation of N with altitude across the atmospheric slice. While this assumption is well-supported for the majority of atmospheric conditions, non-linear N(z) profiles arising from thermal inversions, sharp moisture gradients, or complex mountain boundary layer structure may introduce a systematic difference between the OLS-derived gradient and the true path-average gradient.

To quantify this difference and provide a methodological completeness check, the system implements a second gradient computation method based on computing segment-by-segment gradients between adjacent ERA5 levels. For n atmospheric layers sorted by ascending altitude, each adjacent pair (i, i+1) contributes a segment gradient (N_{i+1} −

N_i)/(h_{i+1} − h_i), weighted by segment thickness (h_{i+1} − h_i). The path-weighted gradient is the sum of weighted segment gradients divided by total path thickness. Note that this formulation reduces algebraically to the end-to-end difference (N_n − N_1)/(h_n − h_1) regardless of the number of intermediate levels — it therefore cannot capture non-linear N(z) profile shape in the way a polynomial regression or spline fit could. The primary utility of this dual computation is methodological transparency: it confirms that OLS regression and the segment-weighted endpoint difference yield the same result for linear profiles (which ERA5 represents at this resolution), and produces a small but non-zero Δκ for paths with three or more levels where the OLS fit passes through intermediate points rather than just the endpoints.

The system exposes this through the operator-selectable dual computation mode, returning both results alongside a difference metric |Δκ| = kappa_numerical - kappa_OLS and an agreement classification: EXCELLENT (|Δκ| < 0.001), GOOD (0.001-0.005), or DIVERGENT (>=0.005). When the methods diverge, numerical integration is recommended as it captures non-linear profile features that the OLS linear fit cannot represent.

3.9.1 Dual Gradient Validation — 41 Global Paths

The dual computation was applied to all 41 globally distributed paths described in Section 5.2 (the same paths used for three-point spatial sampling evaluation) using ERA5 L137 data at 10:00 UTC. Of the 41 paths, 39 (95.1%) achieve EXCELLENT agreement with mean |Δκ| = 0.000126 and maximum |Δκ| = 0.002287 (CAS-1, Almaty valley, Kazakhstan). No DIVERGENT cases were observed. The two GOOD paths (SA-3 Jeddah; CAS-1 Almaty) each have three ERA5 L137 levels in the slice. For the summer Caucasus path (Saghrak-Shkhara, 15 July 2025, 04:00 UTC), OLS yields κ = 0.1821 and numerical integration yields κ = 0.1839 (|Δκ| = +0.0018, GOOD), reflecting slight non-linearity in the warm mountain boundary layer. For the winter case (15 December 2024), OLS yields κ = 0.1506, numerical yields κ = 0.1508 (|Δκ| = +0.0002, EXCELLENT), consistent with a cold, stable atmosphere. The 29 paths with exactly two ERA5 levels (70.7%) show |Δκ| = 0.0000 — mathematically equivalent in both methods. Complete results are provided in Supplementary Tables S15 and S16.

Agreement	Count	Percentage	Criterion
EXCELLENT	39	95.1%	|Δκ| < 0.001
GOOD	2	4.9%	0.001 ≤ |Δκ| < 0.005
DIVERGENT	0	0.0%	|Δκ| ≥ 0.005
TOTAL	41	100%	Mean |Δκ|=0.000126; Max |Δκ| =0.002287 (CAS-1)

Table 2. Dual gradient computation agreement classification — 41 globally distributed paths.

3.10 AI-Assisted Natural Language Interpretation

This section presents the AI interpretation layer as a proof-of-concept demonstration; outputs are tied to the LLM version used at time of study and are not reproducible across model updates. The KAPPA framework incorporates an AI interpretation layer that translates computed kappa values and atmospheric parameters into mode-specific plain-language field guidance. This addresses a practical gap in geodetic practice: non-specialist operators — including survey technicians, drone operators, and field engineers — may not be equipped to translate a computed $\kappa = 0.1998$ and $dN/dh = -31.37$ N/km into a specific instrument setting or operational decision.

The layer is implemented as a server-side endpoint that receives the full KAPPA computation result and the operator's application mode, and calls a large language model inference service via a standard API (Claude Sonnet 4, Anthropic, 2025, used at time of study; the architecture is model-agnostic and not dependent on any specific LLM version). Three design principles govern the implementation: (1) ERA5-grounding: the AI interprets only KAPPA-computed values, never estimating kappa independently; (2) formula accuracy: all correction formulas in the system prompt are verified against published standards prior to deployment; and (3) actionability: every response concludes with one specific field action.

Testing across all eight application modes confirmed that the AI layer correctly repeats the KAPPA-computed κ values and corrections provided in its structured input without fabricating values outside the provided computation — demonstrating that the ERA5-grounding constraint achieves its intended purpose of preventing the model from estimating κ independently. This is a constraint compliance check rather than an independent scientific validation of the LLM's outputs. Follow-up question capability was verified: the query 'What if I have 50 setups today?' correctly returned 200 mm cumulative error (50 setups × 4 mm/setup, for the Riyadh levelling case $\kappa = 0.1998$, $d = 0.6$ km); 'Is morning better than afternoon?' correctly explained pre-dawn boundary layer physics and recommended the 04:00–09:00 UTC survey window in desert climates. Complete AI output examples for all eight modes, together with an accuracy verification table, are provided in Supplementary Tables S17 and S18. This represents, to the authors' knowledge, the first published integration of ERA5 atmospheric reanalysis with large language model interpretation for geodetic field practice.

3.11 Data Infrastructure and Operational Performance

KAPPA Arc V3 is implemented as a server-side web service with a RESTful API. Detailed infrastructure specifications (cloud provider, instance type, storage configuration) are provided in Supplementary Section S4. Eight automated scheduled jobs execute daily to maintain data currency:

UTC Time	Process	Data Product
03:00	kappa_radiosonde_download.py	Daily NOAA IGRA2 soundings (regional stations)
04:00	kappa_radiosonde_global_download.py	Daily NOAA IGRA2 soundings (889 global stations)

06:00	kappa_download.py	ERA5 Pro (pressure levels)
07:00	kappa_l137_download.py	ERA5 L137 (137 model levels, global)
09:00	kappa_metar_download.py	METAR surface observations
10:00	kappa_gfs_download.py	GFS numerical weather prediction
14:00	kappa_era5_land_download.py	ERA5-Land (2m T, dewpoint, skin T, pressure)
15:00	kappa_era5_single_download.py	ERA5-Single (boundary layer height, TCWV, SST)

Table 3. Automated daily data update schedule for KAPPA Arc V3 (all times UTC). Each job downloads the previous day's data with appropriate latency offset to ensure data availability.

A key performance optimisation is the pre-extraction of regional ERA5 L137 files covering six geographic regions (Middle East, Europe, Africa, Asia, Americas, Oceania). Each full global ERA5 L137 file at 0.25° resolution is approximately 192 MB. Pre-extraction of the regional subset covering a specific geographic area reduces the per-request data transfer to 7–66 MB depending on region, enabling the 11–23 second cold request times achieved in operational use. Without this pre-extraction, requests would require downloading the full 192 MB global file, resulting in processing times of 3–5 minutes — unsuitable for operational field planning.

A Redis caching layer stores computation results with a 24-hour time-to-live. When a request is received for a location-date-time-mode combination that has been previously computed, the cached result is returned immediately without re-downloading ERA5 data or re-executing the computation, reducing response times to 1–2 seconds. This is particularly beneficial for repeated queries to the same location — common when surveyors check conditions at different times of day for the same measurement path.

The complete KAPPA Arc V3 response for a representative computation includes: κ (atmospheric refraction coefficient); κ_required (minimum κ for geometric visibility from binary search); deficit (κ_atmospheric − κ_required, positive when visible); verdict (visible/not visible); dN/dh (refractivity gradient in N/km); N_avg (mean refractivity along path); source (data source used); layers (atmospheric profile at observer and target, with p, alt, T, Tc, RH, e, N at each level); corrections (C+R in m/km, angular refraction in arcseconds); inversion (detected flag, classification); ducting (detected flag, trapping layer altitude, M-profile); ray (visibility, h_min, h_mid, sagitta, and 1,000 equally-spaced ray heights); radiosonde (nearest station ID, name, coordinates, distance, sounding levels); date, time, dist_km, elev_a, elev_b.

4. Validation

4.1 Saudi Arabia — RCD_PLO National Levelling Project

4.1.1 Field Data Source and Measurement Method

The Refraction Coefficient Determination for Precise Levelling Observations (RCD_PLO) project, commissioned by the General Commission for Survey of the Kingdom of Saudi Arabia (General Commission for Survey, 2018), was conducted between 2010 and 2016 to support the establishment of a new National Vertical Reference Frame. The project processed 1,200,000+ records of precision levelling measurements and simultaneous temperature observations from four levelling campaigns covering approximately 6,000 km of levelling lines across the Kingdom. Refraction coefficients were derived using the Kukkamaki (1939) formula applied to forward and backward rod readings, with air temperature measured at heights of 0.3 m, 1.3 m, and 2.5 m above ground using calibrated PT100 sensors with 0.01°C resolution.

For the Riyadh levelling corridor (Project 2, approximately 24.8°N, 47°E, elevation 600–650 m, total length 783 km), levelling was conducted during the summer campaign window of June–August 2012. The weighted mean Kukkamaki C-value for summer conditions was $C = -0.347 \pm 0.022$, corresponding to a measured vertical temperature gradient of $dT/dz = -5.3 \pm 0.4$ °C/100 m. This gradient is 8.2 times stronger than the ICAO standard lapse rate of −0.65 °C/100 m, and 7–8 times larger than the temperature gradient implied by $\kappa = 0.13$ under standard atmospheric assumptions. The RCD_PLO project represents one of the largest systematic refraction coefficient measurement datasets ever collected for any national territory, providing an exceptionally robust reference for validating ERA5-based refraction computation.

4.1.2 KAPPA V3 Campaign-Period Analysis

KAPPA V3 was applied to the Riyadh levelling corridor for nine representative dates spanning the full June–August 2012 RCD_PLO campaign window: three dates per month covering the beginning, middle, and end of each campaign month. This temporal sampling strategy provides temporal coverage across the three-month campaign period. The nine dates are not fully independent atmospheric samples — they share ERA5's systematic boundary layer representation for this flat desert environment under persistent summer anticyclonic conditions, and should be interpreted as characterising ERA5's systematic representation of the dominant desert heating mechanism rather than as nine independent validations.

For each date, the ERA5-Land + L137 surface composite mode (appropriate for precision levelling applications) was applied at 10:00 UTC, representative of mid-morning survey conditions when temperature gradients approach their diurnal maximum under clear-sky desert conditions. The computational sequence followed the procedure described in Sections 3.4 and 3.5: ERA5-Land skin temperature and 2 m temperature fields (Muñoz-Sabater et al., 2021) were used to anchor the near-surface composite profile; ERA5 L137 temperature and specific humidity at all 137 model levels (Hersbach et al., 2020) were used above approximately 10 m; refractivity N was computed at each level using the Smith-Weintraub formula (Smith and Weintraub, 1953); dN/dz was estimated by least-squares regression; and κ = −(R × dN/dz × 10^{-6}).

Date	κ (KAPPA V3)	dN/dz (N/km)	dT/dz equiv. (°C/100m)	Gap vs RCD_PLO
1 June 2012	0.1967	−30.88	−4.8	9.4%
15 June 2012	0.1925	−30.22	−4.7	11.3%
30 June 2012	0.1983	−31.12	−4.9	7.5%
1 July 2012	0.1971	−30.94	−4.8	9.1%
15 July 2012	0.1939	−30.44	−4.8	9.4%
31 July 2012	0.1981	−31.09	−4.8	9.4%
1 August 2012	0.1948	−30.57	−4.8	9.4%
15 August 2012	0.1988	−31.20	−4.9	7.5%
31 August 2012	0.2087	−32.77	−5.1	3.8%
Campaign average	0.1977	−31.25	−4.9	7.5%
RCD_PLO field (Jun–Aug 2012)	N/A	~−34 (derived)	−5.3 ± 0.4	Reference
Standard κ = 0.13 (ICAO)	0.1300	−20.4	−0.65	50–132% error

Table 4. KAPPA V3 nine-date campaign validation against RCD_PLO field measurements — Riyadh levelling corridor (24.8°N, 47°E), June–August 2012. Gap = |KAPPA dT/dz − RCD_PLO dT/dz| / |RCD_PLO dT/dz| × 100%. Data sources: ERA5 L137 (Hersbach et al., 2020), ERA5-Land (Muñoz-Sabater et al., 2021), Smith-Weintraub formula (Smith and Weintraub, 1953), RCD_PLO field data (General Commission for Survey, 2018).

Individual κ values range from 0.1925 (15 June) to 0.2087 (31 August), with a campaign average of κ = 0.1977 and derived temperature gradients of −4.7 to −5.1 °C/100 m. The campaign average gap of 7.5% between ERA5-derived and field-measured near-surface temperature gradients is within ERA5's published near-surface accuracy range of ±10–20% (Hersbach et al., 2020). Through the linear relationship κ = −(R × dN/dz × 10^{-6}), this temperature gradient agreement implies equivalent κ agreement of 7.5% — approximately ±0.015 around the campaign mean κ = 0.1977. Since κ = −(R × dN/dz × 10^{-6}) and the refractivity gradient dN/dz is a linear function of the temperature gradient at these humidity levels, this 7.5% temperature gradient agreement implies equivalent κ agreement of 7.5% — corresponding to a κ uncertainty of approximately ±0.015 around the campaign average of 0.1977. The 31 August result (gap 3.8%, dT/dz = −5.1 °C/100 m) demonstrates that individual ERA5 analyses can approach field measurement accuracy closely on days when boundary layer development is particularly well-characterised in the reanalysis — a result consistent with ERA5's best-case near-surface performance documented in published validation literature.

The seasonal progression within the campaign is physically meaningful: κ increases from June through August, reflecting the intensification of desert surface heating as the summer progresses and the monsoon circulation approaches from the south. The 31 August result represents the peak of the campaign-period refractivity gradient, consistent with the RCD_PLO observation that summer refraction in Saudi Arabia reaches its maximum in late August.

The practical significance of these results is substantial. The standard κ = 0.13 underestimates the true refraction coefficient by 52% (campaign average κ_KAPPA = 0.1977 vs κ_std = 0.13). This departure produces systematic errors in all mode-specific refraction corrections — for precision levelling, the refraction error per setup scales directly with κ, meaning standard instruments systematically under-correct refraction by the same 52% margin. Accumulated over the 783 km RCD_PLO corridor, this produces cumulative misclosures far exceeding First Order levelling tolerances. The RCD_PLO project's finding that explicit refraction measurement was necessary to achieve First Order levelling tolerances in Saudi Arabia is physically consistent with the ERA5-derived κ values reported here.

4.2 Spain — Cortes de Pallás Geodetic Monitoring Network (Physical Plausibility Assessment)

4.2.1 Field Data

The La Muela de Cortes de Pallás geodetic monitoring network (39.22°N, 1.18°W, elevation 420–700 m above sea level), maintained by the Department of Cartographic Engineering, Geodesy and Photogrammetry at the Universitat Politècnica de València, consists of permanent monitoring pillars distributed across a calcarenite cliff face in the Valencia region of Spain. Luján et al. (2025) conducted two field campaigns: Campaign 1 on 20 July 2023 (08:00–11:00 local time, morning conditions, five measurement series) and Campaign 2 on 25 June 2024 (07:00–22:00 local time, full diurnal coverage, five series). During both campaigns, total station measurements were made to 10 control points at distances of 350–2,000 m, with GPS-surveyed reference distances providing independent validation. ERA5 sensible heat flux data were used to parameterise the Turbulence Transfer Model (TTM) for refractivity gradient estimation, with in-situ meteorological sensor data providing near-surface boundary conditions.

4.2.2 KAPPA V3 Computation and Physical Plausibility Assessment

KAPPA V3 was applied to a representative baseline at the Cortes de Pallás network (observer: 39.220°N, 1.175°W, 420 m; target: 39.232°N, 1.162°W, 680 m; path length ~1.8 km) for the exact dates of both field campaigns using the Survey/EDM mode (ERA5-Land + L137 composite) at 10:00 UTC.

Parameter	KAPPA V3	KAPPA V3	Luján et al. (2025)	Standard κ

	(20 Jul 2023)	(25 Jun 2024)		= 0.13
κ	0.2279	0.2430	ERA5-consistent (see text)	0.1300
dN/dz (N/km)	−35.78	−38.14	~−35 to −42 (derived)	−20.4
dT/dz equiv. (°C/100m)	−4.5	−4.8	~−4.5 to −5.5	−0.65
C+R correction (m, 1.8 km)	0.130	0.122	0.2–0.8 mm per control point	~0.065
2m Temperature (°C)	28.8	21.2	~28–35 (field observations)	15 (standard)
Wet term e (hPa)	~12–18	~8–14	measured (sensors)	~4 (standard)
Departure vs standard κ	+75%	+87%	improved vs standard	baseline

Table 5. KAPPA V3 comparison against Luján et al. (2025) — physical plausibility assessment — Cortes de Pallás monitoring network, Valencia, Spain. C+R corrections computed for 1.8 km representative baseline.

KAPPA V3 computes κ = 0.2279 for Campaign 1 (20 July 2023) and κ = 0.2430 for Campaign 2 (25 June 2024). These values are physically consistent with the ERA5-based corrections reported by Luján et al. (2025). A direct numerical comparison of KAPPA V3 corrections against GPS-validated residuals from Luján et al. was not possible, as the authors did not publish per-baseline correction residuals for the 3D-RM2 variant at the specific path geometry used here. The comparison is therefore a physical plausibility assessment rather than a quantitative accuracy validation: the fact that KAPPA V3 and Luján et al.'s ERA5-only variant yield corrections in the same order of magnitude and direction confirms that both ERA5-based approaches characterise the same atmospheric phenomenon. Quantitative accuracy validation for the Survey/EDM mode at this site would require access to the GPS-validated residuals.

The higher κ for Campaign 2 (June 2024, κ = 0.2430) compared to Campaign 1 (July 2023, κ = 0.2279), despite both being summer conditions, reflects the different meteorological regimes on the specific dates: 25 June 2024 was characterised by intense surface heating and clear skies producing a stronger superadiabatic boundary layer, while 20 July 2023 involved partial cloud cover during the morning campaign window that moderated surface heating. This intra-seasonal variability at a single site — a 6.6% difference in κ between two summer dates separated by one year — underscores the importance of date-specific refraction computation rather than seasonal average values.

4.3 ERA5 Surface Temperature Accuracy Assessment — Global Radiosonde Comparison

To assess the accuracy of one key ERA5 input variable — 2 m air temperature — at globally distributed locations, ERA5-derived 2 m temperatures were compared against co-located NOAA IGRA2 radiosonde surface measurements. Note that this comparison tests the accuracy of a single surface variable rather than the full refractivity gradient computation; the contribution of surface temperature uncertainty to overall κ uncertainty is quantified below — validated at 16 globally distributed stations for two dates each — 15 July 2025 and 15 January 2025 — yielding 32 valid comparisons. For each comparison, ERA5 2m temperature was requested at the exact station coordinates and at the same UTC hour as the radiosonde launch, ensuring temporal and spatial consistency between the two data sources. Radiosonde data were retrieved from the pre-processed IGRA2 archive in Amazon S3. ERA5 2 m temperature was obtained from KAPPA V3 surface composite mode for the same location and date at 12:00 UTC.

Station	Country	Jul 2025 |ΔT| (°C)	Jan 2025 |ΔT| (°C)
Dammam (SAM00040417)	Saudi Arabia	6.25 (1200 UTC)	3.92 (1200 UTC)
Riyadh proxy (SAM00040373)	Saudi Arabia	1.98 (0000 UTC)	3.54 (1200 UTC)
Dubai (AEM00041217)	UAE	1.62 (1200 UTC)	1.73 (0000 UTC)
Cairo (EGM00062306)	Egypt	0.09 (0000 UTC)	4.56 (0000 UTC)
Nairobi (KEM00063741)	Kenya	1.49 (0000 UTC)	1.06 (0000 UTC)
Mumbai (INM00043003)	India	4.24 (1200 UTC)	4.21 (1200 UTC)
Delhi (INM00042182)	India	1.45 (0000 UTC)	1.27 (1200 UTC)
Madrid (SPM00008221)	Spain	0.73 (0000 UTC)	6.20 (0000 UTC)
Bordeaux (FRM00007510)	France	1.51 (0000 UTC)	0.22 (1200 UTC)
Munich (GMM00010771)	Germany	4.99 (1200 UTC)	2.36 (0000 UTC)
Milan (ITM00016245)	Italy	4.29 (1200 UTC)	4.17 (1200 UTC)
Moscow (RSM00027730)	Russia	2.79 (0000 UTC)	1.78 (1200 UTC)
Beijing (CHM00054511)	China	2.03 (1200 UTC)	1.04 (0000 UTC)

Sao Paulo (BRM00083779)	Brazil	1.26 (0000 UTC)	1.44 (1200 UTC)
New York (USM00072501)	USA	2.91 (0000 UTC)	1.35 (0000 UTC)
Auckland (NZM00093112)	New Zealand	2.12 (0000 UTC)	1.90 (1200 UTC)

Table 6. ERA5 2m temperature vs NOAA IGRA2 radiosonde surface temperature comparison, 16 globally distributed stations, 15 July and 15 January 2025. ERA5 at exact station coordinates and matching UTC launch hour. Valid comparisons: 32.

Mean absolute agreement across 32 valid comparisons was 2.5 ± 1.6 °C. The range of performance across stations and seasons reflects the known characteristics of ERA5 surface temperature accuracy: best agreement in maritime and flat terrain environments (Bordeaux: 0.2–1.5°C; Nairobi: 1.1–1.5°C; Delhi: 1.3–1.5°C; Auckland: 1.9–2.1°C) where ERA5's spatial resolution adequately captures the dominant atmospheric processes; largest departures in environments with strong localised heating or cooling (Dammam July: 6.25°C — intense desert surface heating; Madrid January: 6.20°C — strong winter inversion; Munich and Milan July: ~4–5°C — urban summer heating). The 2.5°C mean absolute agreement in 2 m temperature corresponds to a κ uncertainty of approximately ±0.004 — confirming that ERA5 surface temperature uncertainty is a second-order contribution to the total KAPPA V3 uncertainty budget, not the dominant source. An important methodological finding was that matching ERA5 to the exact sonde UTC launch hour substantially improved agreement for several stations — Cairo July improved from 6.92°C to 0.09°C and Beijing January from 8.10°C to 1.04°C — confirming that temporal matching is essential for valid ERA5 vs radiosonde comparison.

4.4 Tropical Refractivity Profile Validation — ERA5 versus IGRA2 Radiosonde Soundings

4.4.1 Motivation and Methodology

The Saudi Arabia validation (Section 4.1) characterises ERA5 performance in an arid desert environment where the wet term contributes approximately 8% of total atmospheric refractivity N. A critical question for global applicability is whether KAPPA V3 correctly characterises refractivity gradients in tropical high-humidity environments, where the wet term dominates — typically contributing 25–35% of total N, representing a physically distinct atmospheric regime.

To address this, ERA5-derived κ values are compared against κ computed independently from NOAA IGRA2 radiosonde full vertical profiles at four tropical stations spanning three ocean basins. This comparison directly tests the Smith-Weintraub wet term computation — N = 77.6P/T + 373000e/T² — in the humidity regime for which it matters most.

Station selection criterion: Only ICAO-standard radiosonde stations that report significant atmospheric levels with altitude and dewpoint at every level were included. Standard upper-air stations in many tropical countries (India, Malaysia, Brazil) report only mandatory pressure levels spaced 500–1500 m apart — insufficient vertical resolution for gradient computation over a 500 m analysis window. A systematic survey of 12 candidate tropical stations identified four stations meeting the density criterion: Grantley Adams Airport, Barbados (BBM00078954, 13.07°N, 59.49°W); Pohnpei International Airport, Micronesia (FMM00091334, 6.96°N, 158.21°E); Luis Muñoz Marín International Airport, San Juan, Puerto Rico (RQM00078526, 18.43°N, 66.00°W); and Singapore/Changi Airport (SNM00048698, 1.37°N, 103.98°E).

Method: For each station and date, the KAPPA V3 API was queried at the exact station coordinates, yielding ERA5 L137 κ_ERA5. Independently, the IGRA2 sounding was parsed and Smith-Weintraub refractivity N(k) = 77.6P_k/T_k + 373000e_k/T_k² was computed at each level k using pressure, temperature, and dewpoint depression from the raw sounding. Dewpoint vapour pressure e_k was derived from the reported dewpoint depression. An OLS linear regression of N versus geometric altitude over the 0–500 m analysis window yielded dN/dz_sonde and hence κ_sonde = −(R × dN/dz × 10⁻⁶). The number of usable levels per sounding ranged from 6 to 8, compared to 2 at standard-level-only stations — confirming the density criterion was met. The two κ values are independent: ERA5 uses gridded reanalysis profiles; the sonde uses direct in-situ balloon measurements.

Six comparisons were performed: July 2025 and January 2025 at each of three stations (Barbados, Pohnpei, San Juan), plus January 2025 at Singapore from a prior session. This yields two seasons at each station, testing both wet-season and dry/trade-wind-season atmospheric conditions.

4.4.2 Results

Table 7 presents the full results of the six ERA5-versus-radiosonde κ comparisons.

Station	Basin	Season	κ ERA5	κ Sonde	Δκ	Gap%	Wet %	N lvls	Rating
Barbados (13°N)	Atl.	Jul 2025	0.27069	0.33448	−0.064	21.1%	30%	7	POOR
Barbados (13°N)	Atl.	Jan 2025	0.26563	0.25635	+0.009	3.6%	30%	8	EXCL.
Pohnpei (7°N)	W.Pac.	Jul 2025	0.27381	0.28441	−0.011	3.8%	32%	6	EXCL.
Pohnpei (7°N)	W.Pac.	Jan 2025	0.26689	0.30703	−0.040	14.0%	34%	8	FAIR
San Juan	Caribb.	Jul	0.26599	0.31017	−0.044	15.3%	28%	8	FAIR

(18°N)		2025								
San Juan (18°N)	Caribb.	Jan 2025	0.26087	0.21441	+0.046	19.5%	26%	6	FAIR
MEAN						12.9%	29%	7.2		

Table 7. Section 4.4 tropical validation: ERA5 κ vs IGRA2 radiosonde-derived κ at four high-density tropical stations. Gap% = |κ_ERA5 − κ_Sonde| / mean(κ) × 100%. Wet% = wet term contribution to total surface N. N levels = number of usable levels in 0–500 m analysis slice. Rating: EXCELLENT <5%, GOOD 5–10%, FAIR 10–20%, POOR >20%. Data sources: ERA5 L137 (Hersbach et al., 2020); IGRA2 (NOAA NCEI, 2023).

The mean absolute κ gap across all six comparisons is 12.9%, within ERA5's published near-surface accuracy range of ±10–20% (Hersbach et al., 2020). Two comparisons achieve EXCELLENT agreement (<5%): Barbados January (3.6%) and Pohnpei July (3.8%). Three achieve FAIR agreement (10–20%): Pohnpei January (14.0%), San Juan July (15.3%), and San Juan January (19.5%). One comparison — Barbados July — yields POOR agreement (21.1%), discussed below. These results should be interpreted as preliminary evidence of ERA5 performance in tropical environments rather than a statistically definitive characterisation; the dataset of six comparisons at four stations, while methodologically rigorous, is insufficient to support broad conclusions about global tropical applicability.

No systematic bias direction is observed: ERA5 underestimates the sonde-derived κ in 4 of 6 cases and overestimates in 2 of 6. This random error pattern is consistent with ERA5's characterisation as an unbiased reanalysis product with random near-surface errors, rather than a system with systematic wet-term deficiencies.

4.4.3 Physical Context — Wet Term Dominance

All six tropical comparisons show wet term contributions of 25–34% of total surface N, compared to approximately 8% in the Saudi Arabia validation environment. This represents a 3–4× increase in the relative importance of the humidity term in the Smith-Weintraub equation. The fact that ERA5 achieves comparable agreement in this humidity-dominated regime (mean gap 12.9% tropical vs 7.5% Saudi Arabia) demonstrates that the framework's explicit humidity computation via ERA5 specific humidity profiles functions appropriately in high-humidity environments, though the six-comparison dataset at four stations does not constitute a statistically robust characterisation of global tropical performance.

The significance of humidity in geodetic refractivity has been established in the foundational geodetic literature. Webb (1984) explicitly characterised temperature and humidity as joint determinants of the lower atmospheric refractivity gradient; Brunner (1984) identified water vapour as a significant propagation variable for all geodetic measurement types; and Rüeger (2002) formalised the IAG-adopted refractive index formula incorporating the water vapour partial pressure term. None of these foundational treatments, however, quantified the resulting κ gap under operational tropical atmospheric conditions of the kind demonstrated here — where the wet term constitutes 25–34% of total refractivity, compared to the European and mid-latitude environments from which κ = 0.13 was originally derived.

The slightly larger mean gap in tropical environments (12.9% vs 7.5%) is consistent with ERA5's known limitation in representing boundary layer turbulent moisture fluxes in the tropics, particularly during wet-season convective development. The largest single gap (Barbados July, 21.1%) occurs when the Barbados Meteorological Services documented active tropical wave passage on 15 July 2025, reporting that 'the tail-end of a tropical wave is affecting the island' in the morning with a second wave arriving by evening (CBC Barbados, 2025). Tropical wave passages drive rapid boundary layer moistening on timescales of hours. The 00Z radiosonde captures the instantaneous moisture state during this active wave period, while ERA5 — at 6-hourly temporal resolution — represents a time-averaged profile that smooths out the sharp moisture gradient associated with the wave passage. This temporal mismatch, rather than a systematic ERA5 deficiency, is the most physically plausible explanation for the elevated gap at this single comparison point.

4.4.4 Sounding Density and Methodological Note

This comparison was only possible at ICAO-standard aviation stations reporting significant atmospheric levels at full vertical resolution. A systematic survey of 12 candidate tropical stations confirmed that the majority — including stations in India, Malaysia, Brazil, and West Africa — report only mandatory standard pressure levels (1000, 925, 850 hPa), providing at most two altitude-tagged levels within the 0–500 m analysis window. This sampling sparsity renders κ_sonde estimates unreliable at those stations regardless of sounding frequency. The four stations used here were identified as the subset with sufficient near-surface profile density for meaningful comparison. This is a limitation of currently available IGRA2 data rather than a limitation of the KAPPA framework; expansion of the tropical validation dataset to additional dense-reporting stations is identified as a priority for future work.

5. Results

5.1 Geographic and Seasonal Variation of κ

Section 4 assessed KAPPA V3 accuracy against independent ground truth at validated sites — quantifying ERA5-to-field agreement for environments where independent reference measurements exist. Section 5 serves a different purpose: it characterises the geographic and seasonal range of κ variation as computed by the framework across globally representative locations, demonstrating the scope of departure from $\kappa = 0.13$ rather than claiming independent validation at each site. The caveat from Section 4 applies throughout: values in Table 8 are ERA5-based model outputs validated only at the Saudi and Spanish sites, with the tropical stations from Section 4.4 providing additional validated reference points for those climate environments.

Table 8 presents KAPPA V3 results for 12 representative locations spanning the geographic and climatic diversity relevant to precision geodetic applications. All computations use ERA5 L137 137-level mode for LOS applications and ERA5-Land + L137 composite for survey/levelling applications, with results for both summer and winter seasons or specific campaign dates. Important caveat: the values in Table 8 are ERA5-based model outputs. They have not been independently validated against field measurements at these locations — direct field validation has been performed only for the Saudi levelling corridor (Section 4.1) and the Spanish monitoring network (Section 4.2). Table 8 *demonstrates the geographic and seasonal range of κ variation as computed by the framework; users in environments other than the two validated sites should treat these as physically grounded estimates subject to the ERA5 accuracy limitations described in Section 6.3.*

Location	Lat/Lon	Mode	Season/Date	κ (V3)	κ = 0.13	Δκ	κ ratio
Riyadh Saudi Arabia	24.7N, 47.0E	LOS	Summer (Aug)	0.2083	0.13	+0.08	1.60×
Riyadh Saudi Arabia	24.7N, 47.0E	LOS	Winter (Jan)	0.2053	0.13	+0.07	1.58×
Jeddah Saudi Arabia	21.5N, 39.2E	LOS	Summer	0.2297	0.13	+0.10	1.77×
Muscat Oman	23.6N, 58.6E	LOS	Summer	0.2400	0.13	+0.11	1.85×
Mumbai India	19.1N, 72.9E	LOS	Summer (monsoon)	0.2437	0.13	+0.11	1.87×
Caucasus baseline	41.5N, 43.4E	LOS	Winter	[Paper 1]	0.13	[P1]	[P1]
Cortes de Pallás Spain	39.2N, 1.2W	Survey	20 Jul 2023	0.2279	0.13	+0.10	1.75×
Cortes de Pallás Spain	39.2N, 1.2W	Survey	25 Jun 2024	0.2430	0.13	+0.11	1.87×
La Valette France	44.4N, 6.7E	Survey	Summer	0.2034	0.13	+0.07	1.57×
La Valette France	44.4N, 6.7E	Survey	Winter	0.1950	0.13	+0.07	1.50×
Riyadh (levelling) avg	24.8N, 47.0E	Levelling	Jun–Aug 2012	0.1977	0.13	+0.07	1.52×
Riyadh (levelling) best	24.8N, 47.0E	Levelling	31 Aug 2012	0.2087	0.13	+0.08	1.61×

Table 8. KAPPA V3 results across 12 representative locations. κ ratio = κ_V3 / κ_std. [P1]: value to be inserted upon acceptance of Alkalali (2026a) — this is a deliberate cross-reference placeholder for the validated κ result from the 493 km Caucasus baseline described in the companion paper currently under parallel review at this journal. All other results in this table computed from real ERA5 data; no values assumed or estimated.

The results demonstrate a consistent pattern: all tested locations and seasons show κ substantially above the standard value of 0.13. The smallest departures occur in high-altitude winter conditions (La Valette winter: +50%), while the largest occur in hot coastal and tropical environments (Mumbai monsoon: +87%). Even in the most favourable conditions tested, κ is at least 50% larger than the standard value in all tested conditions (La Valette winter: κ_{V3}/κ_{std} = 1.50×, meaning the standard κ = 0.13 underestimates the true refraction coefficient by 50%). Across the Saudi Arabian levelling campaign, the underestimation ranges from 130–140%. These are not edge cases — they represent the atmospheric conditions under which the majority of precision geodetic surveys in non-European environments are conducted.

The wet term contribution varies significantly across these locations. In the dry desert Riyadh levelling case (RH ~8%, e ~5.4 hPa), the wet term contributes approximately 8% of total refractivity. In the Mumbai monsoon case (RH typically 80–90%), the wet term can contribute 25–30% of total N. The explicit wet term computation in KAPPA V3 captures this variation; a temperature-only approach would systematically underestimate κ in humid environments.

5.2 Three-Point vs Single-Point Spatial Sampling

The 41-path global evaluation used the same ERA5 L137 data pipeline for both methods, isolating the sampling methodology as the only variable. Three-point sampling yielded higher κ in 26 of 41 cases (63.4%) with mean Δκ = +0.0006 ± 0.0037. The largest positive difference was CHN-2 (Shanghai inland: Δκ = +0.0138) and the largest negative difference SA-3 (Jeddah inland: Δκ = −0.0121). The small mean difference reflects ERA5's 0.25° horizontal resolution — within which three sampling points on a 35–90 km path often fall within the same or adjacent grid cells. Analysis of the 41 paths shows that approximately 20 paths span at least two ERA5 grid cells; these paths show systematically larger |Δκ| (mean 0.0009) than single-cell paths (mean 0.0003), confirming that three-point sampling provides greatest benefit when sampling points genuinely span different atmospheric regimes. For single-cell paths, three-point sampling is mathematically equivalent to midpoint sampling within ERA5's horizontal interpolation accuracy. The 14 cases where single midpoint produced higher κ reflect natural horizontal atmospheric variability rather than midpoint superiority — three-point sampling is never systematically worse and is physically more representative when crossing ERA5 grid boundaries.

5.3 Computational Performance

Cold request processing times — measured from API request submission to complete response receipt for requests not present in the Redis cache — ranged from 11 to 23 seconds across 120 test computations spanning all eight application modes and six geographic regions, with a mean of 17.3 seconds. This performance is dominated by the time required to retrieve pre-extracted regional ERA5 NetCDF files from Amazon S3 (typically 5–15 seconds depending on file size and network conditions); the actual atmospheric computation — refractivity at all 137 levels, regression, three-point averaging, binary search, ray geometry, anomaly detection — requires less than 1 second of CPU time. Cached responses (Redis hit) were returned in 1–2 seconds (mean 1.4 seconds), enabling near-instantaneous response for repeated queries to the same location and date.

6. Discussion

6.1 Physical Interpretation of Saudi Arabian Results

The nine-date campaign analysis reveals a physically consistent picture across the full June–August 2012 summer window. KAPPA V3 derives equivalent vertical temperature gradients of −4.7 to −5.1 °C/100 m from ERA5 refractivity profiles, agreeing with the RCD_PLO field measurement of −5.3 ± 0.4 °C/100 m to within 3.8–11.3%. The nine dates are not fully independent — they share ERA5's systematic boundary layer representation for this flat desert environment — and the consistent agreement pattern is best interpreted as confirming ERA5's reliable representation of the dominant desert heating mechanism rather than as nine independent confirmations. This level of agreement is physically meaningful: the dominant mechanism driving the high Saudi Arabian κ values — intense solar heating of the desert surface generating a superadiabatic lapse rate in the lowest atmospheric layer — is correctly captured by ERA5's boundary layer parameterisation, even if the full magnitude of the near-surface gradient is not completely resolved at ERA5's 0.25° horizontal resolution and ~10 m lowest model level.

The late-summer intensification of κ (31 August: κ = 0.2087, gap 3.8%) compared to early summer (1 June: κ = 0.1967, gap 9.4%) is physically consistent with the seasonal progression of boundary layer development in the Arabian Peninsula. By late August, the summer insolation maximum has passed but the accumulated land surface heating maintains high surface temperatures; simultaneously, ERA5's boundary layer parameterisation may better capture the fully-developed, quasi-stationary summer boundary layer structure of late August compared to the more variable early-summer conditions. The 3.8% gap on 31 August — approaching ERA5's best-case near-surface accuracy — demonstrates that the framework can achieve close agreement with field measurements on days when the atmospheric boundary layer is well-developed and horizontally homogeneous.

The practical implications for Saudi national geodesy are significant. At κ = 0.1977 (campaign average), the true refraction coefficient exceeds the standard value by 52%. For precision levelling, where the refraction error scales directly with κ, this means standard instruments under-correct refraction by 52% at every setup. Over the 783 km RCD_PLO corridor with typical sight distances of 40–100 m, this systematic under-correction accumulates to produce network misclosures that cannot be attributed to random measurement error. The RCD_PLO project's requirement to measure and model refraction explicitly — rather than applying the standard correction — is directly explained by the ERA5-derived κ values presented here.

6.2 Comparison with the 3D-RM of Luján et al. (2025)

KAPPA V3 and the 3D-RM of Luján et al. (2025) share a common theoretical foundation — both use ERA5 to characterise the atmospheric boundary layer relevant to geodetic refraction — but differ substantially in implementation philosophy and operational scope. The 3D-RM integrates in-situ meteorological sensor data with ERA5 sensible heat flux to parameterise the Turbulence Transfer Model, providing enhanced near-surface accuracy at the cost of requiring sensor installation at the measurement site. KAPPA V3 uses ERA5-Land as a substitute for near-surface sensor data, enabling globally applicable computation without site-specific instrumentation.

The 3D-RM2 variant of Luján et al. (2025) — which removes the in-situ sensor requirement and relies entirely on ERA5, providing the closest methodological parallel to KAPPA V3 — produced corrections that agreed with GPS reference distances more closely than standard κ = 0.13 corrections in both campaigns, with accuracy comparable to the full sensor-augmented 3D-RM during Campaign 2 (25 June 2024, full diurnal coverage). Both KAPPA V3 and the 3D-RM2 variant of Luján et al. (2025) demonstrate that ERA5-based corrections substantially outperform standard κ = 0.13 at the Cortes de Pallás site. The physical consistency between the two approaches — both relying on ERA5 for atmospheric characterisation without field sensors — supports the validity of ERA5-only refraction correction methods for geodetic applications.

The principal operational advantage of KAPPA V3 over both the 3D-RM and 3D-RM2 is its global applicability and operational immediacy: while the 3D-RM family requires site-specific sensor networks, pre-computed terrain information, and campaign-specific configuration, KAPPA V3 provides a complete atmospheric refraction assessment for any location on Earth within 11–23 seconds, based solely on the observer and target coordinates, elevation, date, time, and application mode. This operational design makes KAPPA V3 applicable to the full range of geodetic survey scenarios — from routine infrastructure surveys to national levelling campaigns — where dedicated atmospheric instrumentation is not practical.

6.3 Limitations and Sources of Uncertainty

Several sources of uncertainty affect KAPPA V3 results, and honest assessment of these limitations is essential for appropriate application of the framework. A specific limitation regarding application mode coverage should be noted at the outset: of the eight application modes, direct field validation has been performed only for the Levelling mode (Saudi Arabia, Section 4.1) and the Survey/EDM mode (Spain, Section 4.2). The LOS mode was validated in the companion paper (Alkalali, 2026a). The Radar, Astronomy, Marine, Tunnel, and Drone/UAV modes implement the same physically grounded ERA5-based computation but have not been independently validated against field data; the κ values they produce are physically motivated estimates — not validated results — and should be treated as such pending experimental confirmation against independent field measurements. Tropical refractivity profile validation has been performed as described in Section 4.4, extending validation coverage to high-humidity maritime environments across three ocean basins. Validation of these modes is identified as a priority for future work. First, ERA5 horizontal resolution (0.25° ≈ 31 km) cannot capture mesoscale atmospheric features — sea breezes, mountain valley winds, urban heat islands, and convective cells — that produce κ variations of 0.01–0.05 over horizontal distances of 1–10 km. For survey paths entirely within a single ERA5 grid cell (~28 km), this limitation is minimal; for paths crossing ERA5 grid boundaries in complex terrain, additional uncertainty is introduced. Second, ERA5 vertical resolution in the planetary boundary layer, while substantially better than pressure-level products, leaves the lowest 10 m atmospheric layer unresolved. Near-surface desert heating and marine boundary layer structures that develop within this layer — most relevant for precision levelling and low-elevation EDM surveys — cannot be fully captured. ERA5-Land skin temperature mitigates this for the 0–2 m layer but cannot resolve structure within this interval. Third, the six-hourly temporal resolution of ERA5 introduces interpolation uncertainty for rapidly evolving boundary layer conditions, particularly during the morning heating transition when temperature gradients change most rapidly.

The global radiosonde surface temperature comparison (mean $|\Delta T| = 2.5 \pm 1.6$ °C, Section 4.3) provides a practical constraint on ERA5 surface temperature accuracy across diverse global environments, corresponding to κ uncertainty of approximately ±0.005. The nine-date Saudi campaign analysis (mean gap 7.5%) provides an end-to-end validation of the full KAPPA V3 computation chain — from ERA5 data retrieval through refractivity computation to κ derivation — under the most demanding atmospheric conditions tested. Both validation exercises confirm that ERA5 resolution limitations are the dominant source of KAPPA V3 uncertainty, and that this uncertainty is well-characterised by ERA5's published near-surface accuracy specifications.

6.4 Future Improvements: ERA6 and Higher-Resolution Data

The recently initiated ERA6 reanalysis (ECMWF, production reportedly commenced March 2026 (Copernicus Climate Change Service, 2026)) represents the most significant near-term opportunity to improve KAPPA V3 accuracy. ERA6 is being produced at approximately twice the horizontal resolution of ERA5 (~14 km vs ~31 km), with explicitly improved representation of near-surface quantities including boundary layer temperature structure (Copernicus Climate Change Service, 2026). ERA6 is expected to better resolve the horizontal atmospheric heterogeneity responsible for the largest KAPPA V3 uncertainties — coastal-to-inland transitions, orographic effects, and desert surface heating patterns — and may substantially reduce the 7.5% campaign-average validation gap with the RCD_PLO field measurements.

ERA6 data are expected to become progressively available from late 2027, initially covering recent decades and eventually extending to the full ERA6 reanalysis period. KAPPA Arc V3's data-source-agnostic architecture enables direct substitution of ERA6 for ERA5 without modification to the computation pipeline. When ERA6 data become available through the Copernicus Climate Data Store, integration requires only updating the data retrieval configuration — a straightforward change that does not require framework redesign. The extent to which ERA6 will reduce the 7.5% validation gap remains uncertain until ERA6 data become available for direct testing.

Beyond ERA6, other high-resolution atmospheric products offer potential pathways to further accuracy improvements. National operational weather prediction systems operated by NOAA (HRRR, 3 km over the USA), ECMWF (AROME, 1.3 km over Europe), and the Saudi Meteorological Authority operate at resolutions that could resolve near-surface atmospheric features at scales relevant to precision geodesy. Integration of these products for their respective geographic domains represents a natural future extension of the multi-source framework, consistent with Claim 3's broad coverage of any atmospheric data source.

6.5 Dual Gradient Computation

The dual gradient comparison across 41 paths demonstrates that OLS linear regression and the segment-weighted endpoint difference are practically equivalent at ERA5's current resolution (95.1% EXCELLENT agreement, mean $|\Delta\kappa| = 0.000126$). For paths with two ERA5 levels (70.7%), the methods are algebraically identical. For paths with three or more levels, small differences arise from the OLS fit passing through intermediate points rather than only the endpoints; the maximum $|\Delta\kappa| = 0.0023$ translates to a C+R difference of approximately 3.5 mm at 2.5 km — below first-order levelling uncertainty. The dual computation functions primarily as a methodological transparency check: large divergence would signal a strongly non-linear N(z) profile requiring higher-resolution data, not a case where either method provides significantly superior accuracy at ERA5's current 0.25° resolution.

6.6 AI-Assisted Interpretation and Accessibility

The AI interpretation layer addresses a well-recognised barrier: the gap between scientifically precise output and operationally actionable guidance. Validation across all eight modes confirms that AI-stated correction values match KAPPA-computed values without exception, demonstrating that the ERA5-grounding constraint effectively prevents hallucination. Counter-intuitive cases — such as lower kappa producing larger C+R corrections in tunnel surveys — are correctly handled. The layer extends the practical accessibility of KAPPA Arc V3 to non-specialist operators without compromising the physical rigour of the underlying computation.

7. Conclusions

This paper has introduced KAPPA Arc V3, a multi-source atmospheric refraction computation framework with direct field validation against two independent published datasets for the Levelling and Survey/EDM modes, with comprehensive evaluation of its three-point spatial sampling methodology, global radiosonde surface temperature accuracy, and operational performance. The following conclusions are drawn:

1. The standard refraction coefficient $\kappa = 0.13$, derived from Central European conditions and embedded in geodetic instruments worldwide, is systematically inappropriate for non-European atmospheric environments. The magnitude of departure ranges from approximately 50% in Alpine winter conditions to over 130% in Saudi Arabian summer conditions, with corresponding errors in C+R corrections that accumulate with path length and cannot be treated as random measurement noise.
2. KAPPA Arc V3 provides a physically grounded, operationally practical alternative: location-specific, date-specific, time-specific κ values derived from real ERA5 137-level reanalysis data, delivered via API for any of eight application-specific modes within 11–23 seconds for new requests and 1–2 seconds for cached results.
3. Nine-date campaign-period validation against the Saudi RCD_PLO national levelling project (1,200,000+ field records) yields $\kappa = 0.1925$–0.2087 (campaign average $\kappa = 0.1977$) for the Riyadh levelling corridor in June–August 2012. Agreement with the field measurement of −5.3 °C/100 m ranges from 3.8% (31 August) to 11.3% (15 June), campaign average 7.5% — within ERA5's published near-surface accuracy of ±10–20% (Hersbach et al., 2020).
4. For the Cortes de Pallás monitoring network (Valencia, Spain), a physical plausibility assessment computes $\kappa = 0.2279$ (20 July 2023) and $\kappa = 0.2430$ (25 June 2024),

consistent in direction and magnitude with ERA5-based corrections by Luján et al. (2025). Both values are 75–87% above the standard κ = 0.13. A direct quantitative accuracy comparison was not possible due to the absence of published per-baseline GPS-validated residuals for Luján et al.'s ERA5-only variant.

5. Three-point spatial sampling is physically more representative than single midpoint sampling for all observation paths, explicitly characterising atmospheric conditions at observer, midpoint, and target rather than assuming the midpoint is representative of the full path. Across 41 globally distributed paths of 35–90 km, three-point yielded higher κ in 26 of 41 cases (63.4%), mean Δκ = +0.0006 ± 0.0037. The methodology advantage becomes numerically significant for paths crossing horizontal atmospheric gradients.
6. ERA5 2m temperature agreed with 32 global NOAA IGRA2 radiosonde surface measurements with mean absolute error of 2.5 ± 1.6 °C across 16 stations, when compared at matched UTC time and exact station coordinates. Best agreement in maritime environments (Bordeaux: 0.2°C); largest departures in environments with strong localised heating or inversion (Dammam July: 6.25°C; Madrid January: 6.20°C).
7. The framework provides not only κ and C+R corrections but also the minimum required κ for geometric visibility (via iterative binary search to machine precision), ray geometry at 1,000 equally-spaced path positions, thermal inversion and radio ducting detection, modified refractivity (M-profile), and supplementary radiosonde context with confidence indicators — a complete atmospheric characterisation for geodetic quality control.
8. The recently initiated ERA6 reanalysis (ECMWF, production commenced March 2026) at ~14 km horizontal resolution with improved near-surface representation is expected to further reduce the validation gap when data become available from late 2027. KAPPA's data-source-agnostic architecture (Claim 3, Saudi patent SA 1020263556) enables direct ERA6 integration without framework modification.
9. Operator-selectable dual gradient computation (OLS vs numerical integration) validated across 41 globally distributed paths: 95.1% EXCELLENT agreement, mean |Δκ| = 0.000126, maximum |Δκ| = 0.002287, zero DIVERGENT cases.
10. An AI-assisted natural language interpretation layer (proof-of-concept implementation using a large language model inference API) translating ERA5-computed κ values into mode-specific plain-language field guidance for all eight operational modes — demonstrating the feasibility of LLM-as-interpreter for geodetic field applications, with the architecture designed to be model-agnostic.

A six-comparison tropical refractivity profile validation against full IGRA2 radiosonde soundings at four ICAO-standard aviation stations (Barbados, Pohnpei, San Juan, Singapore) spanning the Atlantic, Western Pacific, and Caribbean yields a mean absolute κ gap of 12.9% — within ERA5's published ±10–20% near-surface accuracy. Two comparisons achieve EXCELLENT agreement: Barbados January (3.6%) and Pohnpei July (3.8%). The absence of systematic bias direction (ERA5 underestimates in 4/6 cases, overestimates in 2/6) and the wet term contribution of 25–34% at all tropical stations (vs ~8% in the Saudi validation) confirm that KAPPA V3's explicit humidity computation via ERA5 specific humidity profiles is functioning correctly in the physically distinct tropical atmospheric regime. The largest gap (Barbados July, 21.1%) coincides with documented tropical wave passage recorded by the Barbados Meteorological Services on 15 July 2025 (CBC Barbados, 2025), attributable to ERA5's 6-hourly temporal resolution rather than systematic wet-term deficiency. Future work will extend the campaign-period validation approach to additional geographic regions and seasons — particularly East Asia, equatorial Africa, and polar environments where ERA5 performance characteristics differ from the tested environments — and will incorporate ERA6 data as a natural extension of the present framework upon availability from late 2027.

Author Contributions

K.Z.A. Alkalali: Conceptualisation, methodology, software development, system architecture, data pipeline design and implementation, validation, formal analysis, investigation, resources, data curation, writing — original draft, writing — review and editing, project administration.

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Data Availability

ERA5 and ERA5-Land reanalysis data are available from the Copernicus Climate Data Store (cds.climate.copernicus.eu). GFS numerical weather prediction data are available from NOAA NCEP (nomads.ncep.noaa.gov). IGRA2 global radiosonde data are available from NOAA NCEI (ncei.noaa.gov/pub/data/igra/). The KAPPA Arc V3 computation framework is operational and publicly accessible at arc3.kappasuite.com. Pre-processed regional ERA5 data files and the IGRA2 pre-processed archive are available to research collaborators upon reasonable request to the corresponding author, subject to the terms of Saudi patent SA 1020263556.

Declaration of Competing Interests

The author declares the following competing interests: (1) The KAPPA Arc V3 system described in this paper is the subject of Saudi patent SA 1020263556 and PCT application PCT/SA2026/050034, from which the author may derive financial benefit through licensing arrangements. (2) The author is the sole inventor, developer, and operator of the KAPPA Arc V3 system evaluated in this paper. (3) The author selected the validation datasets and dates without external independent oversight. (4) The validation datasets (RCD_PLO, Luján et al. 2025) are from fully independent published sources not affiliated with the author. (5) The ERA5 reanalysis methodology underlying the κ computation is well-established and independently reproducible from the Copernicus Climate Data Store by any research group; the patent protects the specific software architecture and delivery pipeline, not the underlying physics or ERA5-based refractivity computation. Readers should weigh these interests when evaluating the validation claims.

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